ECON 0150 | Homework 3.1

Due: Friday, Oct. 23 at 5PM

Homework is designed to both test your knowledge and challenge you to apply familiar concepts in new applications. Answer clearly and completely. You are welcomed and encouraged to work in groups so long as your work is your own. Submit your figures and answers to Gradescope.

Q1. Working with a Normal Distribution

The wait times (in minutes) at a restaurant follow a normal distribution with mean $\mu = 12$ minutes and standard deviation $\sigma = 2.5$ minutes. Start each question with a rough sketch of your answer, then find the number using code. Two useful functions from Exercise 3.1:

```
from scipy import stats
# stats.norm.cdf()
# stats.norm.ppf()
```

a) What is the population mean (μ) wait time? *(warm up question)*

b) What is the population variance of the wait time? *(warm up question)*

c) What is the wait time that is longer than exactly 77 percent of wait times? *(use python)*

d) What is the probability that a wait time will be greater than 10 minutes? *(use python)*

e) What is the probability that the wait time will be between 10 and 14 minutes? *(use python)*

f) What is the probability that the wait time will be less than 7 or greater than 17 minutes? *(use python)*

Q2. Sample vs. Population

We know the restaurant's population parameters: $\mu = 12$ and $\sigma = 2.5$. But in the real world, we don't know the population. We only have a sample. In this question, you'll draw samples from the known population and see how well sample statistics approximate population parameters.

a) Use `np.random.normal(12, 2.5, 10)` to draw a sample of $n = 10$ wait times. Compute the sample mean and sample standard deviation (`np.std(sample, ddof=1)`). How do they compare to μ and σ?

b) Repeat part a) for $n = 100$ and $n = 1000$. How do the sample statistics change as n increases?

c) Plot a histogram of your $n = 1000$ sample. On the same figure, overlay the population PDF using:

```
x = np.linspace(4, 20, 200)
plt.plot(x, stats.norm.pdf(x, 12, 2.5))
```

How well does the sample histogram approximate the population?

d) In your own words, what is the relationship between a sample and a population? Why do larger samples give us more information about the population?